

7 *Use of the Data Booklet is relevant to this question.*

Some airbags in cars become inflated when nitrogen gas is produced by the decomposition of solid sodium azide, NaN_3 , inside the airbag.

Assume that metallic sodium is also produced in the reaction.

An airbag contains 5.00 g of sodium azide which decomposes completely on impact.

What is the volume of nitrogen gas produced, measured at a pressure of $9.85 \times 10^4 \text{ Pa}$ and at a temperature of 30.0°C ?

- A** 0.00295 dm^3 **B** 0.292 dm^3 **C** 1.97 dm^3 **D** 2.95 dm^3